DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

Course: B. Tech.

Branch: Electronics & telecommunication **Semester :VI**

Subject Code & Name:BTETC601 Antennas & wave propogation

Max Marks: 60 **Date:** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

			Marks
Q. 1	**Solve Any Two of the following.**		
A)	Sketch the structure of atmosphere & explain each layer, significance of all		**6**
B)	Define the terms virtual height, MUF & their relevance in wave propagation		**6**
C)	Draw and explain multi hop propagation system		**6**
Q.2	**Solve Any Two of the following.**		
A)	What is fading effect? What are the different types of fading?		**6**
B)	Discuss the analysis of a small dipole antenna		**6**
C)	Derive radiation power density		**6**
Q. 3	**Solve Any One of the following.**		
A)	Discuss antenna parameters		**12**
B)	Calculate the length of dipole ,reflector, director 1, director 2 ,director 3 along with the spacing between element for channel 5 VHF = III the frequency band for Cj channel 5 is 174 MHZ to 181 MHz		**12**
Q.4	**Solve Any Two of the following.**		
A)	Explain broadside array concept		**6**
B)	Compare broadside array & end fire array		**6**
Q. 5	**Solve Any two of the following.**		
A)	Draw & explain V antenna		**6**
B)	Draw & explain helical antenna		**6**
C)	Explain yagi uda antenna		**6**

	***** End *****	

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